Fernando M. Toro

fmt3930@nyu.edu | (407) 965-7077 | LinkedIn

Education

New York University — Brooklyn, NY
Bachelor of Mechanical Engineering — 2023 – Expected 2028
Relevant Coursework: Statics, Dynamics, Strength of Materials, Materials Science, Mechanical Design, Fluid Mechanics

Polk State College — Winter Haven, FL
Associate of Arts Degree — Fall 2023

Skills & Interests

Technical: SolidWorks, Onshape, Revit, VEX Robotics, Arduino (C/C++), Python, 3D Printing, Laser Cutting
Languages: English and Spanish
Interests: Robotics, motion systems, sustainable transportation, mechanical design, product design

Engineering & Design Projects

STEP Vertically Integrated Project – Bipedal Robot Design — Brooklyn, NY
Mechanical Designer — Fall 2025 – Present
• Designed ankle brackets and joint connectors integrating the robot leg and foot while supporting two controlled degrees of freedom.
• Developed a foot-grip structure to improve traction and stability during movement.
• Collaborated with teammates to design and iterate mechanical components in Onshape.

Pneumatic Turbine Power System — Winter Haven, FL
Independent Design Project — Summer 2025 – Present
• Designed and iterated a turbine-based pneumatic engine using Onshape, 3D-printed components, shafts, and rolling-element bearings.
• Developed custom rotor, blade, and housing geometries to improve compressed-air interaction and turbine performance.
• Expanded the system to mechanically drive an electric motor as a generator to supply electrical power to auxiliary components.

Opposed-Cylinder Engine Concept — Winter Haven, FL
Independent Design Project — 2026 – Present
• Developing a compact opposed-cylinder engine architecture focused on piston, connecting-rod, crankshaft, and cylinder geometry.
• Modeled and iterated internal engine components in CAD to evaluate motion, packaging, and mechanical clearances.
• Refining cylinder and crank geometry toward a complete functional mechanical assembly for future analysis and prototyping.

Technical Experience

RunTheCode — Davenport, FL
Robotics & Coding Instructor — Summer 2026
• Taught robotics and engineering fundamentals through hands-on projects using VEX Robotics systems.
• Introduced students to Python programming and basic logic, loops, variables, and problem solving.
• Taught basic Arduino programming using C/C++ and guided students through simple hardware-integration exercises.
• Assisted students with assembling, testing, debugging, and improving robotic systems.

JMT Show Concepts — Winter Haven, FL
Assistant Technician — Spring 2021 – Summer 2023
• Assembled and configured Hippotizer media-server systems used for live concert productions.
• Installed primary and redundant backup servers into portable equipment cases for transport and deployment.
• Assisted senior technicians with troubleshooting connectivity and video-output issues during system setup.